

REST - Get country based on country code

Code:

CountryService.java – Service Package

```
package com.cognizant.spring_learn.service;

import com.cognizant.spring_learn.model.Country;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.ApplicationContext;
import org.springframework.stereotype.Service;

import java.util.ArrayList;
import java.util.List;

@Service
public class CountryService {

    @Autowired
    private ApplicationContext context;

    public Country getCountry(String code) {

        List<Country> countryList = new ArrayList<>();

        countryList.add((Country) context.getBean("india"));
        countryList.add((Country) context.getBean("usa"));
        countryList.add((Country) context.getBean("japan"));
        countryList.add((Country) context.getBean("germany"));

        for (Country country : countryList) {
            if (country.getCode().equalsIgnoreCase(code)) {
                return country;
            }
        }

        return null;
    }
}
```

CountryController.java – Controller Package

```
package com.cognizant.spring_learn.controller;

import com.cognizant.spring_learn.model.Country;
import com.cognizant.spring_learn.service.CountryService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.context.ApplicationContext;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

import java.util.ArrayList;
import java.util.List;

@RestController
public class CountryController {

    @Autowired
    private ApplicationContext context;

    @Autowired
    private CountryService countryService;

    // Returns India's details
    @RequestMapping("/country")
    public Country getCountryIndia() {
        return (Country) context.getBean("india");
    }

    // Returns all countries
    @GetMapping("/countries")
    public List<Country> getAllCountries() {

        List<Country> countryList = new ArrayList<>();

        countryList.add((Country) context.getBean("india"));
        countryList.add((Country) context.getBean("usa"));
        countryList.add((Country) context.getBean("japan"));
        countryList.add((Country) context.getBean("germany"));

        return countryList;
    }

    // Returns a country based on the country code
    @GetMapping("/countries/{code}")
    public Country getCountry(@PathVariable String code) {
        return countryService.getCountry(code);
    }
}
```

Country.java – Model Package

```
package com.cognizant.spring_learn.model;

public class Country {

    private String code;
    private String name;

    public Country() {
    }

    public Country(String code, String name) {
        this.code = code;
        this.name = name;
    }

    public String getCode() {
        return code;
    }

    public void setCode(String code) {
        this.code = code;
    }

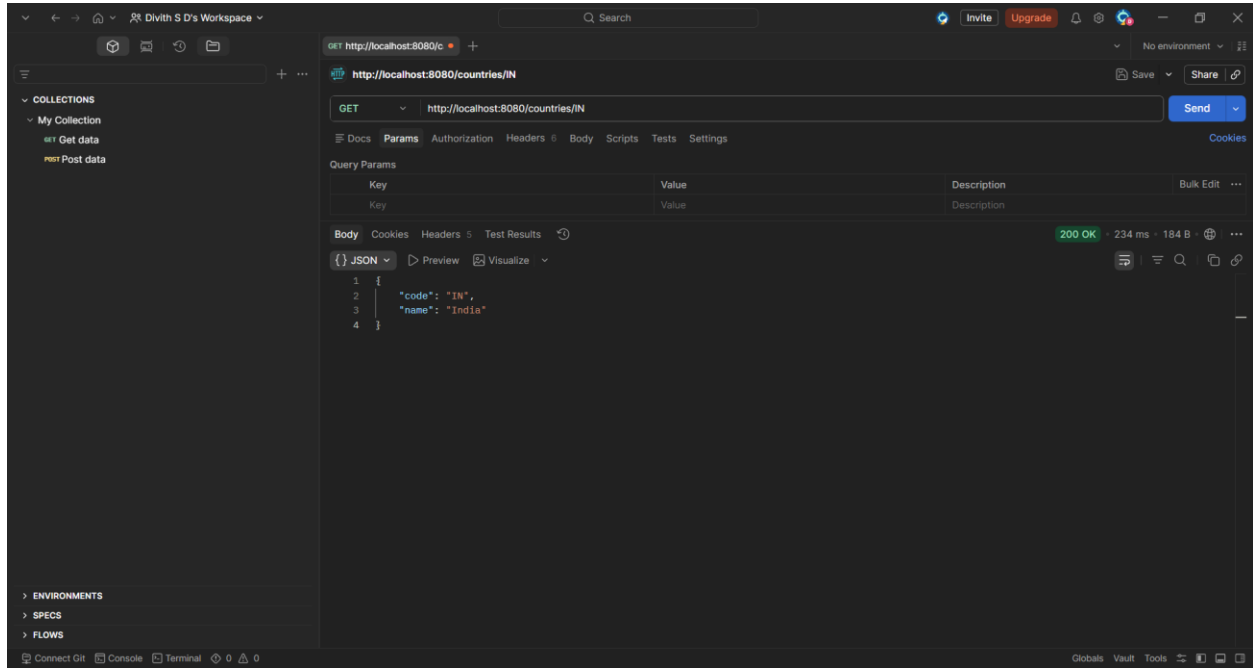
    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}
```

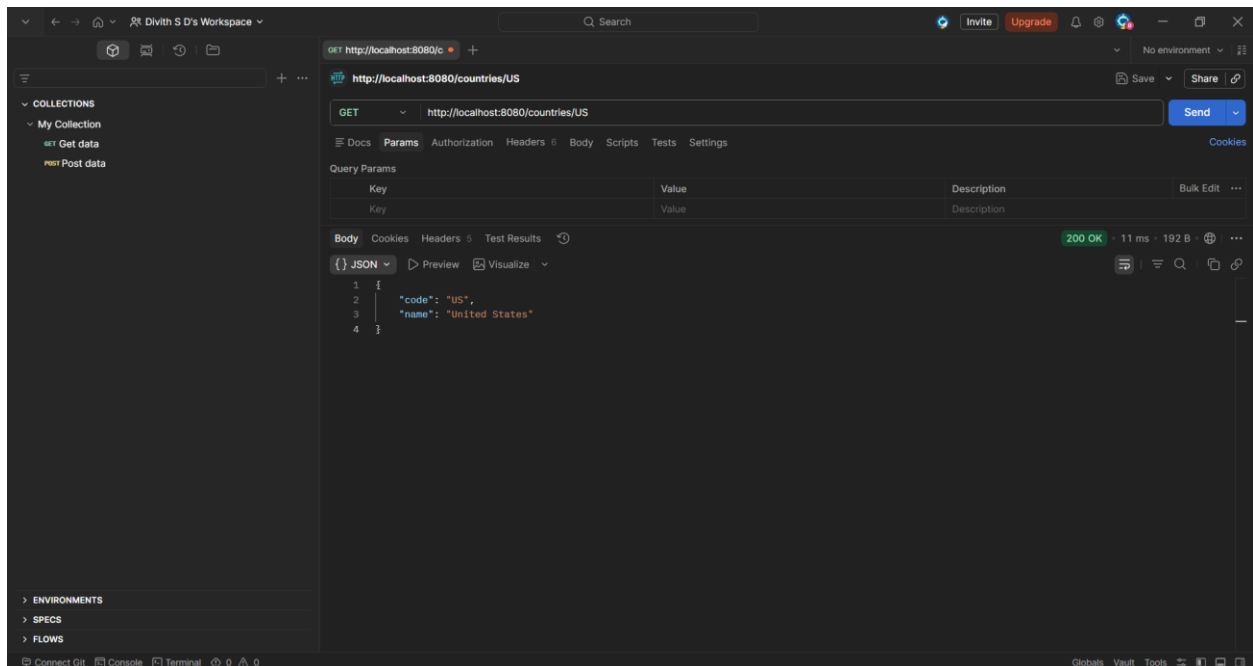
Output:

Postman API Testing:

API Testing - <http://localhost:8080/countries/IN>



API Testing – <http://localhost:8080/countries/US>



API Testing - <http://localhost:8080/countries/JP>

The screenshot shows the Postman interface with a GET request to `http://localhost:8080/countries/JP`. The response is a 200 OK status with a response time of 6 ms and a body size of 184 B. The response body is a JSON object:

```
{  "code": "JP",  "name": "Japan"}
```

The left sidebar shows the 'My Collection' with 'Get data' and 'Post data' requests. The bottom status bar indicates 'Connect Git', 'Console', 'Terminal', and '0 0 0'.

API Testing - <http://localhost:8080/countries/DE>

The screenshot shows the Postman interface with a GET request to `http://localhost:8080/countries/DE`. The response is a 200 OK status with a response time of 6 ms and a body size of 186 B. The response body is a JSON object:

```
{  "code": "DE",  "name": "Germany"}
```

The left sidebar shows the 'My Collection' with 'Get data' and 'Post data' requests. The bottom status bar indicates 'Connect Git', 'Console', 'Terminal', and '0 0 0'.